

 Lecture 2.pdf

read and understand this. dont say anything

 Gemini 3.5 Flash

I have read and fully understood the contents of the document. It covers:

- **Definitions & Proofs:** Independent events ($P(E \cap F) = P(E) \cdot P(F)$), pairwise vs. complete independence, the Law of Total Probability, and Bayes' Theorem.
- **Worked Problems:** Ideal coin tosses (2 and 3 coins), rolling a pair of fair dice, fire engine availability, identifying defective items/parts from factory machines/plants via Bayes' Theorem, drawing balls of different colors from multiple urns, a medical diagnostic scenario for lung cancer, and a principal selection problem featuring total probability and posterior probability calculations.

I am ready to assist you with any questions or tasks you have regarding these concepts or problems.

i will give you topic name or screenshots. you will explain that very clearly, in details, with proper example and intuition. if that is a math , you will state the problem first, then solve it properly. add necessary tables if needed

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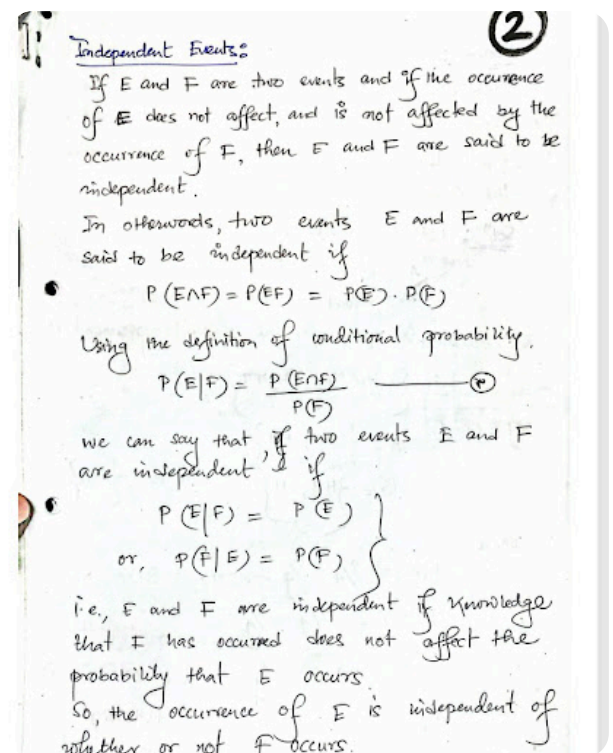
Understood. I'm ready to act as your study companion for this course. Whenever you send over a topic name or a screenshot, I will break it down into four main pillars:

- **Core Intuition:** The "why" behind the concept, explained in plain English, avoiding dry jargon so you actually grasp how it works in the real world.

- **Formal Definitions & Formulas:** Clear, scannable breakdowns of the mathematical framework (using precise notation) so you are completely prepared for exam-style questions.
- **Step-by-Step Solved Examples:** If the topic involves math or probability, I will explicitly state a clear problem first, map out the given variables, and solve it comprehensively using tables, step-by-step equations, or probability trees.
- **Common Pitfalls:** Highlighting tricky areas where students frequently lose marks (such as confusing pairwise independence with complete independence, or misidentifying prior vs. posterior probabilities).

Just send over the first topic or image whenever you're ready, and we will dive right into it! What are we tackling first?

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 Gemini 3.5 Flash

Let's break down this foundational concept from your lecture notes. This page covers **Independent Events** and how they connect to conditional probability.

1. Core Intuition

Imagine rolling a fair six-sided die and flipping a coin at the same time. If the die lands on a 6, does that change the likelihood of the coin landing on Heads? Absolutely not. The coin still has a 50% chance of landing on Heads.

Because the outcome of the die roll provides **zero new information** about the outcome of the coin flip, we say these two events are **independent**. They operate in completely separate worlds.

Conversely, if an event *does* alter the chances of another occurring, they are **dependent**. For example, if you draw a card from a deck and don't replace it, the probabilities for your second draw completely change.

2. Formal Definitions & Formulas

Mathematically, we define independence in two perfectly equivalent ways based on your notes:

The Product Rule (Multiplication Rule)

Two events E and F are independent if and only if the probability of both happening at the same time (their intersection) is the product of their individual probabilities: $P(E \cap F) = P(E) \cdot P(F)$

The Conditional Probability Rule

By definition, the conditional probability of E given that F has already happened is: $P(E|F) = \frac{P(E \cap F)}{P(F)}$

If E and F are independent, knowing that F occurred gives us no advantage, meaning the conditional probability collapses back into the prior probability: $P(E|F) = P(E)$ or similarly, $P(F|E) = P(F)$

Key Takeaway: E and F are independent if knowing that F has occurred does not affect the probability that E occurs.

3. Step-by-Step Solved Example

Let's look at a concrete mathematical problem to see how we prove independence using the formulas from your notes.

Problem Statement

A single fair six-sided die is rolled once. Let event E be rolling an **even number**, and let event F be rolling a **multiple of 3**. Check whether events E and F are independent or not.

Step 1: Define the Sample Space and Events

- **Sample Space (S):** $\{1, 2, 3, 4, 5, 6\}$ (Total outcomes = 6)
- **Event E (Even number):** $\{2, 4, 6\}$

- **Event F (Multiple of 3):** $\{3, 6\}$
- **Intersection ($E \cap F$, Even AND Multiple of 3):** $\{6\}$

Step 2: Calculate Individual Probabilities

- $P(E) = \frac{\text{Number of even numbers}}{\text{Total outcomes}} = \frac{3}{6} = \frac{1}{2}$
- $P(F) = \frac{\text{Number of multiples of 3}}{\text{Total outcomes}} = \frac{2}{6} = \frac{1}{3}$
- $P(E \cap F) = \frac{\text{Number of outcomes containing both}}{\text{Total outcomes}} = \frac{1}{6}$

Step 3: Test for Independence using the Product Rule

Let's see if $P(E) \cdot P(F)$ equals $P(E \cap F)$: $P(E) \cdot P(F) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$

Since $P(E \cap F) = \frac{1}{6}$ and $P(E) \cdot P(F) = \frac{1}{6}$, they match perfectly!

Because $P(E \cap F) = P(E) \cdot P(F)$, events E and F are independent.

4. Common Pitfalls

The "Mutually Exclusive" Trap ❌

The single biggest mistake students make in exams is confusing **independent events** with **mutually exclusive (disjoint) events**.

- **Mutually Exclusive** means the events **cannot happen at the same time** ($P(E \cap F) = 0$). If one happens, the other *cannot* happen. (e.g., turning left vs. turning right).
- **Independent** means one happening **tells you nothing** about the other ($P(E \cap F) = P(E) \cdot P(F)$).

In fact, if two events have non-zero probabilities, they *cannot* be both mutually exclusive and independent! If they are mutually exclusive, knowing F happened means the chance of E happening drops instantly to 0—which means they strongly depend on each other.

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You Asked



Here is the detailed breakdown and explanation of the two coin-tossing problems provided in the screenshots, focusing on the core concept of testing for independence.

Problem 1: Two-Coin Toss

Problem Statement

Two ideal coins are tossed. Let A be the event "head on the first coin" and B be the event "head on the second coin". Check whether the events are independent or not.

Core Intuition

If you flip two separate coins, what happens to the first coin physically has no impact on what happens to the second coin. To prove this mathematically, we test if the joint probability of getting heads on both coins equals the product of their individual probabilities.

Step-by-Step Solution

Step 1: Define the Sample Space (S) When flipping two coins, there are four equally likely outcomes: $S = \{HH, HT, TH, TT\}$

Step 2: Identify Outcomes for Events A , B , and their Intersection ($A \cap B$)

- **Event A (Head on 1st coin):** The first letter must be 'H'. $A = \{HH, HT\}$
- **Event B (Head on 2nd coin):** The second letter must be 'H'. $B = \{HH, TH\}$
- **Intersection ($A \cap B$, Head on both):** The outcome present in both sets. $A \cap B = \{HH\}$

Step 3: Calculate the Probabilities

- Total number of outcomes in sample space $|S| = 4$.
- $P(A) = \frac{|A|}{|S|} = \frac{2}{4} = \frac{1}{2}$
- $P(B) = \frac{|B|}{|S|} = \frac{2}{4} = \frac{1}{2}$
- $P(A \cap B) = \frac{|A \cap B|}{|S|} = \frac{1}{4}$

Step 4: Verify Independence via Product Rule Multiply the individual probabilities: $P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

Comparing this to our intersection probability: $P(A \cap B) = \frac{1}{4} = P(A) \cdot P(B)$

Conclusion: Since $P(A \cap B) = P(A) \cdot P(B)$, the events A and B are **independent**.

Problem 2: Three-Coin Toss

Problem Statement

Three coins are tossed. Show that the events A ("heads on the first coin") and B ("tails on the last two coins") are independent.

Core Intuition

Even when dealing with three coins, knowing that the first coin landed on Heads should tell you absolutely nothing about the behavior of the remaining two coins. We track all 8 possibilities to formally establish this.

Step-by-Step Solution

Step 1: Define the Sample Space (S) A clean way to map three coins is using a sample space grid matching the handwriting:

First \ Last Two	HH	HT	TH	TT
H	HHH	HHT	HTH	HTT
T	THH	THT	TTH	TTT

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$ Total outcomes $|S| = 8$.

Step 2: Identify Outcomes for Events A , B , and $A \cap B$

- **Event A (Heads on the first coin):** Look at the top row of our grid. $A = \{HHH, HHT, HTH, HTT\}$
- **Event B (Tails on the last two coins):** Look for outcomes ending in 'TT'. $B = \{HTT, TTT\}$
- **Intersection ($A \cap B$, Heads on 1st AND Tails on last two):** $A \cap B = \{HTT\}$

Step 3: Calculate the Probabilities

- $P(A) = \frac{|A|}{|S|} = \frac{4}{8} = \frac{1}{2}$
- $P(B) = \frac{|B|}{|S|} = \frac{2}{8} = \frac{1}{4}$
- $P(A \cap B) = \frac{|A \cap B|}{|S|} = \frac{1}{8}$

Step 4: Verify Independence $P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$

Since $P(A \cap B) = P(A) \cdot P(B) = \frac{1}{8}$, the condition holds.

Conclusion: Events A and B are completely **independent**.

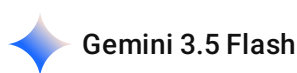
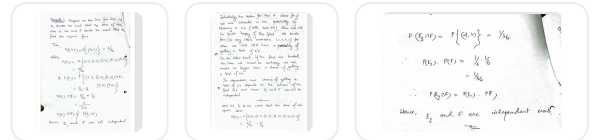
Common Pitfalls to Watch For in Exams

- **Counting Mistakes in B :** For the three-coin problem, students often assume "tails on the last two coins" means *exactly* two tails, forgetting that TTT satisfies the condition perfectly because the final two positions are indeed tails.

- **Incorrect Sample Space Sizes:** Always check your total combinations using 2^n (where n is the number of coins). Two coins = $2^2 = 4$ outcomes ; three coins = $2^3 = 8$ outcomes. If your denominator is wrong, your checks for independence will fail.

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You Asked



Here is a detailed breakdown of the two dice problems from your lecture notes. These examples are classic university exam favorites because they demonstrate how two scenarios that look almost identical on paper can have completely opposite mathematical behaviors regarding independence.

1. Core Intuition

When rolling two dice, the total sample space is always 36 outcomes. Let's look at what independence actually feels like here:

- **The "Sum of 6" Scenario (Dependent):** Before any dice are rolled, you want a sum of 6. If the first die lands on a 4, you are happy because a sum of 6 is still perfectly possible (if the second die lands on 2). But what if the first die lands on a 6? Suddenly, it is physically impossible to get a sum of 6, since the minimum value of the second die is 1. Because the outcome of the first die heavily changes your chances of getting a total sum of 6, these two events are **dependent**.
- **The "Sum of 7" Scenario (Independent):** A total sum of 7 is uniquely magical. No matter what number the first die lands on (1, 2, 3, 4, 5, or 6), there is always exactly **one** matching number on the second die that can bring the total to 7. Because knowing the value of the first die provides absolutely no advantage or disadvantage toward hitting a total of 7, these events are **independent**.

2. Mathematical Framework

To analyze these problems cleanly, we use a standard 6×6 grid representing the outcomes of two rolled dice:

Die 1 \ Die 2	1	2	3	4	5	6
1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

3. Step-by-Step Solved Problems

Case 1: Testing Sum of 6 (E_1) and First Die is 4 (F)

Problem Statement

Suppose we toss two fair dice. Let E_1 denote the event that the sum of the dice is six and F denote the event that the first die equals four. Check whether E_1 and F are independent.

Step-by-Step Solution

1. Define Event Sets:

- E_1 (Sum is 6) = $\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\} \rightarrow 5$ outcomes.
- F (1st die is 4) = $\{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\} \rightarrow 6$ outcomes.
- Intersection $E_1 \cap F = \{(4, 2)\} \rightarrow 1$ outcome.

2. Calculate Probabilities:

- $P(E_1) = \frac{5}{36}$
- $P(F) = \frac{6}{36} = \frac{1}{6}$
- $P(E_1 \cap F) = \frac{1}{36}$

3. Check Product Rule: $P(E_1) \cdot P(F) = \frac{5}{36} \cdot \frac{1}{6} = \frac{5}{216}$

4. Conclusion: Since $P(E_1) \cdot P(F) \neq P(E_1 \cap F)$ (because $\frac{5}{216} \neq \frac{1}{36}$), E_1 and F are NOT independent.

Case 2: Testing Sum of 7 (E_2) and First Die is 4 (F)

Problem Statement

Let E_2 be the event that the sum of the dice equals seven and F be the event that the first die equals four. Check whether E_2 and F are independent.

Step-by-Step Solution

1. Define Event Sets:

- E_2 (Sum is 7) = $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\} \rightarrow 6$ outcomes.
- F (1st die is 4) = $\{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6)\} \rightarrow 6$ outcomes.
- Intersection $E_2 \cap F = \{(4, 3)\} \rightarrow 1$ outcome.

2. Calculate Probabilities:

- $P(E_2) = \frac{6}{36} = \frac{1}{6}$
- $P(F) = \frac{6}{36} = \frac{1}{6}$
- $P(E_2 \cap F) = \frac{1}{36}$

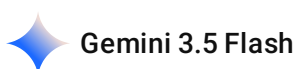
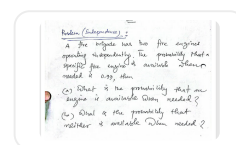
3. **Check Product Rule:** $P(E_2) \cdot P(F) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$

4. **Conclusion:** Since $P(E_2 \cap F) = P(E_2) \cdot P(F) = \frac{1}{36}$, E_2 and F are independent events.

4. Common Pitfalls

- **Blind Intuition Assumptions:** Students often assume that since both problems use the exact same event F ("first die is 4"), their answers must be the same. Always run the product test formula explicitly rather than guessing by gut feeling.
- **Miscounting the Diagonal Outcomes:** When listing out outcomes that sum up to a number, it's easy to miss a pair or accidentally write an invalid pair like $(0, 6)$ or $(7, 0)$. Remember, dice values are strictly integers ranging from 1 to 6.

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Here is the breakdown of the Fire Engine availability problem from your lecture notes. This problem applies the **Addition Rule**, **Complement Rule**, and the property of **Independence** to a real-world

1. Core Intuition

Think of this as a system backup problem. A city has two fire engines working completely independently. If one breaks down or is busy at a scene, it does not affect the functionality of the other.

- **"An engine is available" (Part a):** This implies that the town is safe because **at least one** engine is ready to deploy. It will only fail if *both* crash at the same time. Because we have a backup system, the probability of having at least one engine available (0.9999) is significantly higher than a single engine's individual availability (0.99).
 - **"Neither is available" (Part b):** This is the ultimate failure state where the fire brigade cannot respond because **both** engines are unavailable simultaneously. Because they are independent, this catastrophic overlap is incredibly rare (0.0001).
-

2. Formal Definitions & Framework

To evaluate system reliability with overlapping events, a **Venn Diagram** is the best way to visualize how the individual probabilities interact.

Core Formulas Used:

1. **Product Rule for Independence:** If A and B are independent, the probability of both happening together is: $P(A \cap B) = P(A) \cdot P(B)$
 2. **The Addition Rule (Inclusion-Exclusion Principle):** To find the probability of event A **OR** event B occurring: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 3. **De Morgan's Laws & Complementary Probability:** The probability that neither occurs is the complement of at least one occurring: $P(\overline{A} \cap \overline{B}) = P((A \cup B)^C) = 1 - P(A \cup B)$
-

3. Step-by-Step Solved Problem

Problem Statement

A fire brigade has two fire engines operating independently. The probability that a specific fire engine is available when needed is 0.99. (a) What is the probability that an engine is available when needed? (b) What is the probability that neither is available when needed?

Setup Variables

- Let A = First engine is available $\rightarrow P(A) = 0.99$.

- Let B = Second engine is available $\rightarrow P(B) = 0.99$.
 - Since they operate independently, the probability that both are available is: $P(A \cap B) = P(A) \cdot P(B) = 0.99 \times 0.99 = 0.9801$
-

Solution for Part (a)

Question: What is the probability that **an engine** (meaning engine 1 OR engine 2 OR both) is available?

Using the general addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $P(A \cup B) = 0.99 + 0.99 - 0.9801$
 $P(A \cup B) = 0.9999$

Answer: The probability that at least one engine is available when needed is 0.9999 (or 99.99%).

Solution for Part (b)

Question: What is the probability that **neither** engine is available?

Method 1: Using the Complement of Part (a) The event "neither is available" is the exact opposite of "at least one is available". $P(\overline{A \cap B}) = 1 - P(A \cap B)$
 $P(\overline{A \cap B}) = 1 - 0.9801 = 0.0199$

Method 2: Directly via Independence of Complements If two events are independent, their complements are also independent.

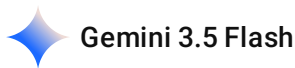
- Probability that engine 1 is unavailable: $P(\overline{A}) = 1 - 0.99 = 0.01$
- Probability that engine 2 is unavailable: $P(\overline{B}) = 1 - 0.99 = 0.01$

Since they fail independently, multiply their failure probabilities: $P(\overline{A} \cap \overline{B}) = P(\overline{A}) \cdot P(\overline{B}) = 0.01 \times 0.01 = 0.0001$

Answer: The probability that neither engine is available is 0.0001 (or 0.01%).

4. Common Pitfalls to Avoid in Exams

- **The Double-Counting Trap** ❌: In part (a), students often simply add $P(A) + P(B) = 0.99 + 0.99 = 1.98$. A probability value can *never* exceed 1.0. You must remember to subtract the overlapping intersection $P(A \cap B)$ because both engines being functional at the same time is counted twice otherwise.
- **Misinterpreting "An Engine"** ❌: Do not confuse "an engine is available" with "exactly one engine is available". "An engine" means the system is operational—whether it's one or both running, the emergency call gets handled. Thus, it requires the Union rule ($\cup$).



This section of your lecture notes handles one of the most subtle and conceptually tricky topics in probability theory: **Pairwise Independence vs. Complete (Mutual) Independence**.

1. Core Intuition

When dealing with two events, independence is straightforward: if A happens, it doesn't change the probability of B . But when we introduce a third event, C , things get interesting.

- **Pairwise Independence:** This means if you pick *any two events* out of the group, they are completely independent of each other. For example, A tells you nothing about B , B tells you nothing about C , and A tells you nothing about C .
- **Complete Independence:** This is a much stricter requirement. It means that even if you know the outcomes of **both** A and B together, you still have absolutely no information about B 's or C 's behavior.

Think of it like a three-way gear lock system. If you look at any two gears individually, they might look like they can turn freely on their own (pairwise independent). But when you try to move all three together, the collective structure restricts them (not completely independent).

2. Formal Definitions & Formulas

For a set of three events A , B , and C to be **completely independent**, they must satisfy two distinct tiers of conditions:

Tier 1: Pairwise Independence (Checked 2 at a time)

Every possible pair must satisfy the standard multiplication rule:

1. $P(A \cap B) = P(A) \cdot P(B)$
2. $P(A \cap C) = P(A) \cdot P(C)$
3. $P(B \cap C) = P(B) \cdot P(C)$

Tier 2: Joint Independence (Checked 3 at a time)

The joint intersection of all three events must also equal the product of all three individual probabilities: $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Definition: If an ensemble of events satisfies Tier 1 but **fails** Tier 2, they are called **pairwise independent but not completely independent**.

3. Step-by-Step Solved Problem

Problem Statement

Two fair coins are tossed. Let A be the event "head on the first coin", B be the event "head on the second coin", and C be the event "the coins fall alike" (both heads or both tails). Show that the events A , B , and C are pairwise independent but not completely independent.

Step 1: Write down the Sample Space (S)

Tossing two coins gives four equally likely outcomes: $S = \{HH, HT, TH, TT\}$ Total outcomes $|S| = 4$.

Step 2: Define the Outcomes for each Event

- **Event A (Head on 1st coin):** First coin must be H. $A = \{HH, HT\}$
- **Event B (Head on 2nd coin):** Second coin must be H. $B = \{HH, TH\}$
- **Event C (Coins fall alike):** Both match (HH or TT). $C = \{HH, TT\}$

Step 3: Find Intersections

Now let's find where these sets overlap with each other:

- **Pairs:**
 - $A \cap B = \{HH\}$
 - $A \cap C = \{HH\}$
 - $B \cap C = \{HH\}$
- **All Three together:**
 - $A \cap B \cap C = \{HH\}$

Step 4: Calculate All Probabilities

Using our counts relative to $|S| = 4$, we evaluate the individual and joint probabilities:

Probability Type	Exact Formula Evaluation	Result
Individual	$P(A) = \frac{2}{4}, P(B) = \frac{2}{4}, P(C) = \frac{2}{4}$	$P(A) = P(B) = P(C) = \frac{1}{2}$
Pairwise Overlaps	$P(A \cap B) = \frac{1}{4}, P(A \cap C) = \frac{1}{4}, P(B \cap C) = \frac{1}{4}$	All equal $\frac{1}{4}$
Joint Overlap	$P(A \cap B \cap C) = \frac{1}{4}$	Equal to $\frac{1}{4}$

Step 5: Test Tier 1 (Pairwise Independence)

Let's multiply the individual probabilities for each pair and check if they match our table values:

- $P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(A \cap B)$ ✓
- $P(B) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(B \cap C)$ ✓
- $P(A) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(A \cap C)$ ✓

Result: The events are perfectly **pairwise independent**.

Step 6: Test Tier 2 (Complete Independence)

Now let's check the product rule for all three events together: $P(A) \cdot P(B) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$

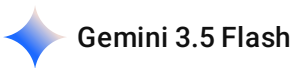
However, looking back at our actual joint intersection calculation from Step 4: $P(A \cap B \cap C) = \frac{1}{4}$

Since $\frac{1}{4} \neq \frac{1}{8}$, we have: $P(A \cap B \cap C) \neq P(A) \cdot P(B) \cdot P(C)$

Conclusion: The events are **pairwise independent but NOT completely independent**.

4. Common Pitfalls

- **The Shortcut Trap** ✗: Assuming that checking $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$ is enough to declare complete independence. It is **not**. You must satisfy *both* the pairwise equations (Tier 1) and the joint equation (Tier 2) simultaneously. It's mathematically possible for the three-way product rule to hold true while some pairs fail!
- **Intuitive Confusion** ✗: Forgetting what event *C* really means. If you know that event *A* happened (Heads on coin 1) and event *B* happened (Heads on coin 2), then event *C* (coins fall alike) is suddenly **100% guaranteed** to happen. This shows that *C* depends heavily on the combined knowledge of *A* and *B*, destroying complete independence.



This section of your notes tackles a highly conceptual and subtly tricky topic in probability theory:
Pairwise Independence vs. Complete (Mutual) Independence.

1. Core Intuition

When we deal with only two events, independence is straightforward: if one happens, it doesn't change the probability of the other. However, when we introduce three or more events, checking independence becomes a multi-layered process.

- **Pairwise Independence:** This means that if you isolate **any two events** out of the group, they have absolutely no effect on each other. For instance, event *A* tells you nothing about event *B*, and event *B* tells you nothing about event *C*.
- **Complete Independence:** This is a much stricter requirement. It means that even if you know the outcomes of **multiple events combined** (e.g., you know *both A and B* happened), that collective knowledge still gives you zero clues about whether event *C* will happen.

The Structural Analogy

Think of a three-way combination lock. If you look at any two gears independently, they might appear to turn freely on their own (pairwise independent). But when you try to turn all three gears simultaneously, the internal mechanism locks them into a shared constraint (not completely independent).

2. Formal Definitions & Formulas

For a set of three events *A*, *B*, and *C* to be classified as **completely independent**, they must satisfy two distinct mathematical tiers:

Tier 1: Pairwise Independence (Checked 2 at a time)

Every single pairs combination must satisfy the basic multiplication rule:

- $P(A \cap B) = P(A) \cdot P(B)$
- $P(B \cap C) = P(B) \cdot P(C)$

$$3. P(A \cap C) = P(A) \cdot P(C)$$

Tier 2: Joint Independence (Checked altogether)

The intersection of all three events combined must equal the product of all three individual probabilities: $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Definition: If a group of events satisfies all conditions in **Tier 1** but fails **Tier 2**, we state that the events are **pairwise independent but not completely independent**.

3. Step-by-Step Solved Problem

Problem Statement

Two coins are tossed. Let A be the event "head on the first coin", B be the event "head on the second coin", and C be the event "coins fall alike". Show that the events A , B , and C are pairwise independent but not completely independent.

Step 1: State the Sample Space (S)

Tossing two fair coins yields four equally likely outcomes: $S = \{HH, HT, TH, TT\}$ Total number of outcomes $|S| = 4$.

Step 2: Define the Sets for Each Event

- **Event A (Head on 1st coin):** Look for outcomes where the first letter is 'H'. $A = \{HH, HT\}$
- **Event B (Head on 2nd coin):** Look for outcomes where the second letter is 'H'. $B = \{HH, TH\}$
- **Event C (Coins fall alike):** Both match (HH or TT). $C = \{HH, TT\}$

Step 3: Identify the Overlapping Intersections

Now, find where these event sets intersect:

- **Pairwise Intersections:**
 - $A \cap B = \{HH\}$
 - $A \cap C = \{HH\}$
 - $B \cap C = \{HH\}$
- **Joint Intersection (All three):**
 - $A \cap B \cap C = \{HH\}$

Step 4: Calculate All Associated Probabilities

Using the counts from our sets relative to the total sample space size (4), we evaluate the probabilities:

Event / Overlap Type	Elements Count	Calculated Probability
Individual Events	\$	A
Pairwise Overlaps	\$	$A \cap B$
Joint Overlap	\$	$A \cap B \cap C$

Step 5: Test for Pairwise Independence (Tier 1)

Let's verify if the multiplication rule holds for every individual pair by multiplying their individual probabilities:

- $P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(A \cap B)$ ✓
- $P(B) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(B \cap C)$ ✓
- $P(A) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(A \cap C)$ ✓

Because every isolated pair satisfies the product formula, the events are completely **pairwise independent**.

Step 6: Test for Complete Independence (Tier 2)

Now, let's check the product rule for all three events taken altogether: $P(A) \cdot P(B) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$

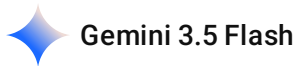
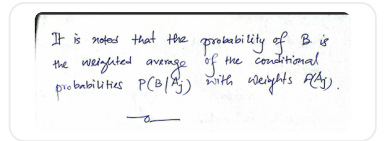
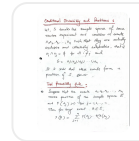
However, looking back at our actual joint intersection calculation from Step 4: $P(A \cap B \cap C) = \frac{1}{4}$

Since $\frac{1}{4} \neq \frac{1}{8}$, we find that: $P(A \cap B \cap C) \neq P(A) \cdot P(B) \cdot P(C)$

Final Conclusion: Because it passes the pairwise check but fails the joint multiplication check, the events *A*, *B*, and *C* are **pairwise independent but not completely independent**.

4. Common Pitfalls to Watch For

- **The Single-Equation Shortcut Trap** ✗: A classic exam mistake is assuming that if $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$ works out, the events are automatically independent. It is a trap! Complete independence requires **all 4 equations** (the 3 pairwise rules + the 1 joint rule) to be true simultaneously.
- **Relying Purely on Casual Intuition** ✗: Looking at the problem statement, you might think, *"Flipping a coin has no memory, so how can they be dependent?"* This shows why checking the math matters. If you are given the information that event *A* happened (Heads on coin 1) and event *B* happened (Heads on coin 2), then event *C* (coins fall alike) is suddenly **100% guaranteed** to happen (*HH*). The combined knowledge completely destroys the independence of *C*.



Gemini 3.5 Flash

This section of your lecture notes covers a fundamental concept in probability: **Sample Space Partitions** and the **Total Probability Rule** (often called the Law of Total Probability).

1. Core Intuition

What is a Partition?

Imagine a complete puzzle spread out on a table. A **partition** simply means slicing the entire puzzle into separate, non-overlapping pieces.

1. **Mutually Exclusive:** No two puzzle pieces can occupy the same space ($A_i \cap A_j = \emptyset$).
2. **Collectively Exhaustive:** If you put all the pieces back together, they perfectly reform the entire puzzle box ($S = A_1 \cup A_2 \cup \dots \cup A_n$).

The Total Probability Rule

Now imagine you spill a drop of water (Event B) across that puzzle. To find the total area of the water drop, you could look at each puzzle piece individually, measure the tiny puddle on that specific piece ($A_j \cap B$), and add them up.

In the real world, we use this when we want to find the overall probability of an outcome (B), but that outcome depends on which hidden scenario (A_j) happens first. It acts as a **weighted average** where each scenario's conditional probability is weighted by how likely that scenario is to occur ($P(A_j)$).

2. Formal Definitions & Formulas

Sample Space Partition

A collection of events $A_1, A_2, \dots, A_n$ forms a partition of the sample space S if:

- $A_i \cap A_j = \emptyset$ for all $i \neq j$ (Disjoint/Mutually Exclusive).
- $\sum_{j=1}^n P(A_j) = 1$ or $S = \bigcup_{j=1}^n A_j$ (Collectively Exhaustive).

Total Probability Rule

Given a partition where $P(A_j) > 0$, the probability of any event $B \in S$ is given by: $P(B) = \sum_{j=1}^n P(A_j) \cdot P(B|A_j)$

3. Step-by-Step Derivation

Let's mathematically trace the proof shown in your notes (Image 2):

- Express B using the partition pieces:** Since the set S is composed of all A_j combined, any event B can be written as its intersection with the whole sample space: $B = S \cap B = (A_1 \cup A_2 \cup \dots \cup A_n) \cap B$. Applying the distributive law yields: $B = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B)$
 - Apply Axiom of Additivity:** Because the chunks $(A_j \cap B)$ are mutually exclusive, we can turn the set unions ($\cup$) into a clean addition ($\sum$) of individual probabilities: $P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B)$ $P(B) = \sum_{j=1}^n P(A_j \cap B)$
 - Substitute with the Conditional Multiplication Rule:** From the definition of conditional probability, we know that $P(A_j \cap B) = P(A_j) \cdot P(B|A_j)$. Plugging this into our summation yields the final formula: $P(B) = \sum_{j=1}^n P(A_j) \cdot P(B|A_j)$
-

4. Step-by-Step Solved Example

Problem Statement

A factory uses three assembly lines (A_1, A_2, A_3) to manufacture microchips. Line A_1 produces 50% of the chips, Line A_2 produces 30%, and Line A_3 produces 20%. The probability of a line producing a defective chip (B) is:

- 1% for Line A_1
- 2% for Line A_2
- 5% for Line A_3

If a chip is selected at random from the final batch, what is the total probability that it is defective?

Step 1: Identify and Organize the Given Variables

First, write down the prior probabilities of a chip coming from each line (our partition):

- $P(A_1) = 0.50$
- $P(A_2) = 0.30$
- $P(A_3) = 0.20$

Next, list the conditional probabilities of getting a defective chip (B) given each line:

- $P(B|A_1) = 0.01$
- $P(B|A_2) = 0.02$
- $P(B|A_3) = 0.05$

Step 2: Set Up a Calculation Table

A matrix style table keeps calculations structured and prevents arithmetic mistakes:

Assembly Line (A_j)	Prior Probability $P(A_j)$	Conditional Defect Rate $P(B A_j)$	Joint Defect Contribution $P(A_j) \cdot P(B A_j)$
Line 1 (A_1)	0.50	0.01	$0.50 \times 0.01 = 0.005$
Line 2 (A_2)	0.30	0.02	$0.30 \times 0.02 = 0.006$
Line 3 (A_3)	0.20	0.05	$0.20 \times 0.05 = 0.010$

Step 3: Sum the Joint Contributions

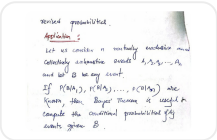
Using the Total Probability formula: $P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$ $P(B) = 0.005 + 0.006 + 0.010 = 0.021$

Answer: The total probability that a randomly chosen microchip is defective is 0.021 (or 2.1%).

5. Common Pitfalls

- **Partition Sanity Check Failure** ❌: Before performing calculations, always make sure the probabilities of your baseline partitioning events sum up to exactly 1.0 ($0.50 + 0.30 + 0.20 = 1.0$). If they don't, they do not form a true partition of the sample space.
- **Swapping $P(B|A_j)$ with $P(A_j|B)$** ❌: Pay extreme attention to the wording in word problems. "The probability that a chip is defective *given it came from Line 1*" is $P(B|A_1)$. Do not mistake it for $P(A_1|B)$, which is the inverse probability handled later by Bayes' Theorem.

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This section of your lecture notes introduces the core philosophy and setup behind **Bayes' Theorem**. It deals with updating your beliefs when new evidence comes to light.

1. Core Intuition

Bayes' Theorem is essentially a structured way to do **data-driven detective work**.

Imagine you wake up with a sore throat. Before looking at any medical data, you might think there is a low baseline chance (1%) that you have a specific rare illness—this initial estimate is your **Prior Probability**.

Then, you take a rapid diagnostic test, and it comes back positive. This test result is your **Additional Information**. Given this new piece of evidence, your 1% initial estimate is no longer accurate; you need to update it to a higher value. This updated, realistic calculation is your **Posterior Probability** (the probability *after* seeing the evidence).

2. Formal Framework & Notation

To use Bayes' Theorem, your system must be partitioned into baseline causes (A_j) and an observable symptom or event (B):

- **The Causes** ($A_1, A_2, \dots, A_n$): These are a set of mutually exclusive and collectively exhaustive events. This means only one cause can be true at a time, and the list covers all possible causes.
- **The Evidence** (B): An event that has occurred or been observed.

The Probability Shifts:

- **Prior Probability** ($P(A_j)$): The initial probability estimate of a specific cause A_j before observing evidence B .
 - **Likelihood** ($P(B|A_j)$): The probability that the evidence B will show up *if* the specific cause A_j is true.
 - **Posterior Probability** ($P(A_j|B)$): The revised probability of the cause A_j being true *given* that we have witnessed evidence B .
-

3. How the Application Works

When you know the baseline rates of your causes ($P(A_j)$) and how likely each cause is to show evidence ($P(B|A_j)$), Bayes' Theorem lets you flip the conditional probability around to find $P(A_j|B)$.

Mathematically, it bridges the gap like this:

$$P(A_i|B) = \frac{\text{Prior Probability} \times \text{Likelihood}}{\text{Total Probability of the Evidence}}$$

Where the total probability of B is found using the partition rule: $P(B) = \sum_{j=1}^n P(A_j)P(B|A_j)$

4. Conceptual Example

Problem Statement

A cyber security tracking system flags an alert (B). The alert can be triggered by two mutually exclusive scenarios: a real Network Hack (A_1) or a System Glitch (A_2).

- Statistically, hacks are rare: $P(A_1) = 0.05$, while glitches are common: $P(A_2) = 0.95$.
- If there is a real Hack, the system alerts 90% of the time: $P(B|A_1) = 0.90$.
- If it's just a Glitch, the system falsely alerts 10% of the time: $P(B|A_2) = 0.10$.

Given that an alert (B) has gone off, what is the posterior probability that a real Network Hack (A_1) is underway?

Step 1: Track the Components

- Prior Probabilities:** $P(A_1) = 0.05$ (Hack), $P(A_2) = 0.95$ (Glitch)
- Likelihoods:** $P(B|A_1) = 0.90$, $P(B|A_2) = 0.10$

Step 2: Calculate Total Probability of the Evidence $P(B)$

Before evaluating the specific cause, find the overall probability of getting an alert using the Total Probability Rule: $P(B) = P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2)$
 $P(B) = (0.05 \times 0.90) + (0.95 \times 0.10)$
 $P(B) = 0.045 + 0.095 = 0.140$

Step 3: Apply Bayes' Theorem to find $P(A_1|B)$

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(B)} = \frac{0.045}{0.140} = 0.3214$$

Answer: Despite the alarm ringing, there is only a 32.14% posterior probability that a real hack is happening because the prior baseline for hacks is initially so tiny.

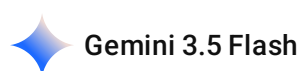
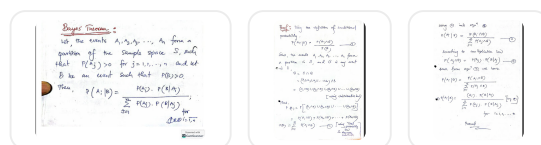
5. Common Pitfalls

- The Conditional Inversion Error** ($P(A|B) \neq P(B|A)$) ❌: Confusing the likelihood with the posterior. For example, "the probability of having a cough if you have flu" ($P(\text{Cough}|\text{Flu})$) is very high, but "the probability of having flu if you have a cough" ($P(\text{Flu}|\text{Cough})$) is quite low

because many minor things cause a cough. Always establish clearly what is the *known event* (B) vs. the *hypothesized cause* (A).

- **Ignoring the Base Rate / Prior** ❌: Forgetting to incorporate the prior probability $P(A_j)$ into your final calculation. A test can be 99% accurate, but if the condition you are testing for is a 1-in-a-million rarity, a positive result still most likely indicates a false positive.

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This section of your lecture notes presents the formal mathematical statement and full algebraic proof of **Bayes' Theorem**. Let's break it down comprehensively, explaining the underlying mechanism of the proof and working through a concrete example.

1. Core Intuition

At its heart, Bayes' Theorem answers a simple question: **"Given that an observed effect has occurred, what is the probability that a specific hidden cause was responsible?"**

Think of the sample space partition ($A_1, A_2, \dots, A_n$) as a set of distinct, non-overlapping pathways. Together, these pathways cover every single way an event can happen. When event B finally occurs, it cuts across these pathways.

Bayes' Theorem isolates the specific slice of event B that belongs to path A_i and divides it by the total size of event B across *all* pathways. It is simply calculating:

Posterior Probability of Cause $A_i = \frac{\text{Size of the overlap between Cause } A_i \text{ and Effect } B}{\text{Total size of Effect } B \text{ across all possible causes}}$

2. Formal Statement & Formula

Let the events $A_1, A_2, A_3, \dots, A_n$ form a partition of the sample space S , such that $P(A_j) > 0$ for $j = 1, 2, \dots, n$, and let B be an event such that $P(B) > 0$.

Then, for any specific cause A_i where $i = 1, 2, \dots, n$, the conditional probability is given by:

$$P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{\sum_{j=1}^n P(A_j) \cdot P(B|A_j)}$$

3. Step-by-Step Mathematical Derivation

Let's break down the proof exactly as written across the second and third pages of your notes.

Step 1: Definition of Conditional Probability

By the fundamental definition of conditional probability, the chance of A_i happening given that B has already occurred is: $P(A_i|B) = \frac{P(A_i \cap B)}{P(B)}$ ——— Equation (1)

Step 2: Breaking Down the Total Denominator, $P(B)$

Since the events $A_1, A_2, \dots, A_n$ partition the sample space S , the event B can be written as its intersection with S : $B = S \cap B$
 $B = (A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n) \cap B$

Applying the **distributive law** allows us to distribute B into each partition piece: $B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B) \cup \dots \cup (A_n \cap B)$

Step 3: Integrating the Law of Total Probability

Because the partitioned regions are mutually exclusive (they do not overlap), the probability of their union is simply the sum of their individual probabilities: $P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B)$
 $P(B) = \sum_{j=1}^n P(A_j \cap B)$ ——— Equation (2)

Step 4: Substituting Equation (2) into Equation (1)

Replacing the denominator $P(B)$ in our original formula gives us: $P(A_i|B) = \frac{P(A_i \cap B)}{\sum_{j=1}^n P(A_j \cap B)}$ ——— Equation (3)

Step 5: Applying the Multiplicative Law

According to the multiplication rule of conditional probability, any joint intersection can be expressed as: $P(A_j \cap B) = P(A_j) \cdot P(B|A_j)$ ——— Equation (4)

Plugging Equation (4) into both the numerator and the denominator of Equation (3) delivers the final proved theorem: $P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{\sum_{j=1}^n P(A_j) \cdot P(B|A_j)}$ (Proved)

4. Step-by-Step Solved Example

Problem Statement

Three automated email servers (A_1, A_2, A_3) route messages for a department.

- Server A_1 handles 40% of traffic, A_2 handles 35%, and A_3 handles 25%.

- The probability of a server misrouting an email (Event B) is 1% for A_1 , 2% for A_2 , and 4% for A_3 .

An email is discovered to have been misrouted (B). Find the probability that it was handled by server A_2 .

Step 1: State the Given Variables

- **Priors:** $P(A_1) = 0.40$, $P(A_2) = 0.35$, $P(A_3) = 0.25$
- **Likelihoods:** $P(B|A_1) = 0.01$, $P(B|A_2) = 0.02$, $P(B|A_3) = 0.04$

Step 2: Set Up the Joint Intersection Contributions Matrix

To make calculation clean and readable, let's track the product terms:

Server (A_j)	Prior $P(A_j)$	Likelihood $P(B A_j)$	Joint Product $P(A_j) \cdot P(B A_j)$
Server 1 (A_1)	0.40	0.01	$0.40 \times 0.01 = 0.004$
Server 2 (A_2)	0.35	0.02	$0.35 \times 0.02 = 0.007$ (Target Numerator)
Server 3 (A_3)	0.25	0.04	$0.25 \times 0.04 = 0.010$
Total Denominator $P(B)$	—	—	$\sum = 0.004 + 0.007 + 0.010 = \mathbf{0.021}$

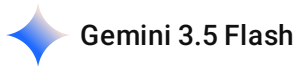
Step 3: Apply Bayes' Theorem Formula

We want to find $P(A_2|B)$: $P(A_2|B) = \frac{P(A_2) \cdot P(B|A_2)}{\sum_{j=1}^3 P(A_j) \cdot P(B|A_j)} P(A_2|B) = \frac{0.007}{0.021} = \frac{1}{3} \approx 0.3333$

Answer: There is a 33.33% probability that the misrouted email came from server A_2 .

5. Common Pitfalls

- **Algebraic Summation Blindness** ❌: In exams, students frequently try to cancel out terms between the numerator and the denominator, like crossing out $P(A_i)$ from the top and bottom. Remember, the denominator is inside a **summation** ($\sum$). You cannot cancel individual items when addition signs link them below the fraction line.
- **Incorrectly Extracting the Likelihood Term** ❌: Be vigilant when reading word problems. "The probability that server 2 misroutes an email" is $P(B|A_2)$. If you accidentally write this value down as $P(A_2|B)$, your entire calculation grid will end up fundamentally scrambled. Always match the conditional bar carefully with your chronological timeline.



Gemini 3.5 Flash

This problem from your notes illustrates an essential, practical application of **Bayes' Theorem**: determining the responsibility of a specific source when a failure or defect occurs.

1. Core Intuition

Think of this problem like a detective investigating a crime scene. A defective item has been found on the factory floor. We have three suspects: Machine 1 (M_1), Machine 2 (M_2), and Machine 3 (M_3).

- **Before looking at the evidence:** If you randomly picked any item from the batch blindly, it was most likely made by Machine 3 because M_3 produces the lion's share (40%) of the factory's inventory. These baseline distributions are your **prior probabilities**.
- **After seeing the evidence:** We now discover that the chosen item is **defective**. This piece of evidence changes everything. Because Machine 1 has the worst track record (5% defect rate) and Machine 3 is highly reliable (2% defect rate), the blame shifts dramatically.

Bayes' Theorem recalculates these distributions to give us the **posterior probability**—the updated probability after accounting for the evidence.

2. Formal Definitions & Formulas

To solve this, we rely on two key mathematical rules:

Bayes' Theorem for Multiple Assemblies

Given a partition of causes $A_1, A_2, \dots, A_n$ and an observed event B , the updated probability of cause A_i is:
$$P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{\sum_{j=1}^n P(A_j) \cdot P(B|A_j)}$$

Conditional Disjoint Union Rule

Because an item cannot be physically manufactured by two machines at the same time, the events A_2 and A_3 are mutually exclusive (disjoint). Therefore, their conditional union simplifies to addition:
$$P(A_2 \cup A_3|B) = P(A_2|B) + P(A_3|B)$$

3. Step-by-Step Solved Problem

Problem Statement

In a factory, three different machines M_1 , M_2 , and M_3 are used for producing a large batch of similar manufactured items. Suppose that M_1 , M_2 , and M_3 produce respectively 25%, 35%, and 40% of the total items. Suppose further that of their items, respectively 5%, 4%, and 2% are defective. Finally, suppose that one item is selected at random from the entire batch and it is found to be defective.

(a) Determine the probability that this item is produced by machine M_1 . (b) Determine the probability that this item is produced by machine M_2 or M_3 .

Step 1: Define Events and Extract Values

- Let A_1 = Item produced by $M_1 \implies P(A_1) = \frac{25}{100} = 0.25$
- Let A_2 = Item produced by $M_2 \implies P(A_2) = \frac{35}{100} = 0.35$
- Let A_3 = Item produced by $M_3 \implies P(A_3) = \frac{40}{100} = 0.40$
- Let B = The item is defective

Now list the conditional defect rates (likelihoods) given by each machine:

- $P(B|A_1) = \frac{5}{100} = 0.05$
- $P(B|A_2) = \frac{4}{100} = 0.04$
- $P(B|A_3) = \frac{2}{100} = 0.02$

Step 2: Create a Joint Probability Contribution Table

We compute the total probability of finding a defective item ($P(B)$) by summing up the individual joint contributions:

Machine (A_j)	Prior $P(A_j)$	Defect Likelihood $P(B A_j)$	Joint Product $P(A_j) \cdot P(B A_j)$
Machine 1 (A_1)	0.25	0.05	$0.25 \times 0.05 = \mathbf{0.0125}$
Machine 2 (A_2)	0.35	0.04	$0.35 \times 0.04 = \mathbf{0.0140}$
Machine 3 (A_3)	0.40	0.02	$0.40 \times 0.02 = \mathbf{0.0080}$
Total Defective $P(B)$	—	—	$\sum = 0.0125 + 0.0140 + 0.0080 = \mathbf{0.0345}$

Solution for Part (a)

We need to find the probability that the defective item came from machine M_1 ($P(A_1|B)$).

Using Bayes' Theorem: $P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(B)}$ $P(A_1|B) = \frac{0.25 \times 0.05}{0.0345} = \frac{0.0125}{0.0345} \approx 0.3623 \rightarrow 0.36$

Interpretation: Approximately 36% of all defective items in this batch are produced by machine M_1 . Notice how its probability grew from a baseline of 25% up to 36% because of its high failure rate.

Solution for Part (b)

We need to find the probability that the defective item was produced by machine M_2 **OR** machine M_3 ($P(A_2 \cup A_3|B)$).

Since A_2 and A_3 are mutually exclusive, we can directly sum their joint components over the total defective denominator: $P(A_2 \cup A_3|B) = \frac{P(A_2) \cdot P(B|A_2) + P(A_3) \cdot P(B|A_3)}{P(B)}$ $P(A_2 \cup A_3|B) = \frac{0.0140 + 0.0080}{0.0345} = \frac{0.0220}{0.0345} \approx 0.6377 \rightarrow 0.64$

Interpretation: There is a 64% probability that a randomly chosen defective item was manufactured by either M_2 or M_3 .

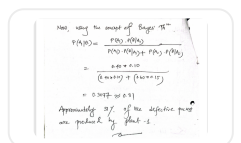
Alternative Shortcut: Since the item *must* come from M_1 , M_2 , or M_3 , you can also find this via the complement rule: $1 - P(A_1|B) = 1 - 0.36 = 0.64$.

4. Common Pitfalls to Watch For

- **Misinterpreting Part (b) as a Simple Union ($P(A_2 \cup A_3)$)** ❌: A very common exam mistake is calculating the baseline production union $P(A_2) + P(A_3) = 0.35 + 0.40 = 0.75$. Remember, you are calculating the conditional union *given that it is already known to be defective* ($|B$). You must evaluate it using the posterior weights.
- **Rounding Intermediate Steps Early** ❌: If you round your denominator 0.0345 or individual products too early to one or two decimal places, truncation errors will propagate, and your final answers can deviate heavily from the true value. Keep at least 4 decimal places during intermediate steps.

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You Asked 



This problem from your lecture notes demonstrates how to use **Bayes' Theorem** to find the probability that a specific manufacturing plant is responsible for a defective batch.

1. Core Intuition

Imagine a company that receives shipments of supply packets from two different factories: Plant 1 and Plant 2.

- **Prior Baseline:** Plant 2 is larger and supplies more packets (60%) than Plant 1 (40%). If you picked a packet completely at random, it would more likely be from Plant 2.
- **The Twist:** Plant 2 has a higher defect rate (15%) compared to Plant 1 (10%).

When a packet is discovered to be **defective**, this piece of evidence alters our suspicion. Even though Plant 2 makes more total packets, it also makes a much higher percentage of junk. Bayes' Theorem precisely weighs these two competing factors—total volume vs. defect rate—to calculate the exact updated probability.

2. Formal Definitions & Formulas

Bayes' Theorem for Two Categories

When the sample space is partitioned into exactly two baseline categories (A_1 and A_2), the posterior probability of category A_1 given evidence B simplifies to:

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2)}$$

3. Step-by-Step Solved Problem

Problem Statement

A company produces sphere parts and supplies them in packets. The company produces 2,000 packets by Plant 1 and 3,000 packets by Plant 2. Previous experience indicates that 10% produced by Plant 1 are defective, while 15% produced by Plant 2 are defective. One day a defective packet was identified. Compute the probability that the packet was produced by Plant 1.

Step 1: Define Events and Convert to Decimals

- A_1 : Packet is from Plant 1 $\rightarrow P(A_1) = \frac{2,000}{2,000+3,000} = \frac{2,000}{5,000} = 0.40$
- A_2 : Packet is from Plant 2 $\rightarrow P(A_2) = \frac{3,000}{2,000+3,000} = \frac{3,000}{5,000} = 0.60$
- B : The randomly selected packet is defective

Now write out the known conditional probabilities (likelihoods):

- $P(B|A_1) = 0.10$ (10% defect rate from Plant 1)
- $P(B|A_2) = 0.15$ (15% defect rate from Plant 2)

Step 2: Set Up the Joint Contribution Table

Source Plant (A_j)	Prior Probability $P(A_j)$	Defect Likelihood $P(B A_j)$	Joint Product $P(A_j) \cdot P(B A_j)$
Plant 1 (A_1)	0.40	0.10	$0.40 \times 0.10 = \mathbf{0.040}$ (Target Numerator)
Plant 2 (A_2)	0.60	0.15	$0.60 \times 0.15 = \mathbf{0.090}$
Total Defective $P(B)$	—	—	$\sum = 0.040 + 0.090 = \mathbf{0.130}$

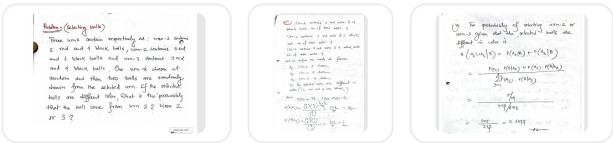
Step 3: Compute the Posterior Probability

Using the Bayes' Theorem formula: $P(A_1|B) = \frac{0.40 \times 0.10}{(0.40 \times 0.10) + (0.60 \times 0.15)} P(A_1|B) = \frac{0.040}{0.130} \approx 0.3077$

Answer: The probability is approximately 0.31 (or 31%), meaning about 31% of the identified defective packets originate from Plant 1.

4. Common Pitfalls

- **Denominator Summation Blunder** ❌: A very easy mistake to make under exam pressure is adding the two baseline defect percentages directly: $P(B) = 0.10 + 0.15 = 0.25$. This completely ignores the fact that the plants produce different total quantities of items. Always compute the full products ($P(A_j) \cdot P(B|A_j)$) before adding them together.
- **Mixing Up the Conditionals** ❌: Make sure you don't confuse $P(B|A_1)$ with $P(A_1|B)$. The problem gives you $P(B|A_1) = 0.10$ (probability that a packet is bad *given* it came from Plant 1). Your goal is to compute the reverse: $P(A_1|B)$ (probability that it came from Plant 1 *given* it is bad).



This problem from your lecture notes demonstrates how to combine **combinatorics (combinations)** with **Bayes' Theorem** to find the posterior probability across a disjoint union of multiple groups.

1. Core Intuition

Imagine standing in front of three distinct, closed urns containing different mixtures of red and black balls.

- **The Process:** First, you blindly pick one of the three urns at random ($1/3$ chance for each). Then, you reach inside and grab exactly two balls.
- **The Evidence:** You open your hand and find that the two drawn balls are of **different colors** (one red and one black).

This specific combination (one red, one black) is much easier to pull out of some urns than others due to the underlying ratios. For example, Urn 2 contains mostly red balls (3 red, 1 black), so pulling out a balanced pair is a different likelihood than pulling from Urn 1 or Urn 3. Bayes' Theorem uses this outcome to update our initial baseline suspicions and evaluate the likelihood that the source was either Urn 2 or Urn 3.

2. Formal Definitions & Formulas

Combinations Rule ($\binom{n}{r}$)

When drawing balls simultaneously from a single container without replacement, order does not matter. We calculate the number of ways to choose r objects from a total pool of n objects using combinations: $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Conditional Probability for the Union of Causes

Since you can only choose one physical urn at a time, selecting Urn 2 (A_2) and selecting Urn 3 (A_3) are mutually exclusive events. Therefore, their posterior joint union simplifies to direct addition:

$$P(A_2 \cup A_3 | B) = P(A_2 | B) + P(A_3 | B) = \frac{P(A_2)P(B|A_2) + P(A_3)P(B|A_3)}{P(B)}$$

3. Step-by-Step Solved Problem

Problem Statement

Three urns contain respectively as: Urn 1 contains 2 red and 4 black balls; Urn 2 contains 3 red and 1 black balls; and Urn 3 contains 3 red and 4 black balls. One urn is chosen at random and then two balls are randomly drawn from the selected urn. If the selected balls are of different colors, what is the probability that the balls came from Urn 2 or Urn 3?

Step 1: Characterize Urn Inventories and Define Events

- **Urn 1 (A_1):** 2 Red + 4 Black = 6 Total balls
- **Urn 2 (A_2):** 3 Red + 1 Black = 4 Total balls
- **Urn 3 (A_3):** 3 Red + 4 Black = 7 Total balls
- Let A_1, A_2, A_3 be the events of picking Urn 1, 2, or 3 respectively. Since selection is completely random: $P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}$
- Let B be the event that the two selected balls are different in color (1 Red and 1 Black).

Step 2: Calculate Likelihoods ($P(B|A_j)$) using Combinations

For each urn, we calculate the number of ways to draw 1 red out of the available red pool **AND** 1 black out of the available black pool, divided by the total possible ways to draw any 2 balls from that urn.

- **Likelihood from Urn 1:** $P(B|A_1) = \frac{\binom{2}{1}\binom{4}{1}}{\binom{6}{2}} = \frac{2 \times 4}{15} = \frac{8}{15}$
- **Likelihood from Urn 2:** $P(B|A_2) = \frac{\binom{3}{1}\binom{1}{1}}{\binom{4}{2}} = \frac{3 \times 1}{6} = \frac{3}{6} = \frac{1}{2}$
- **Likelihood from Urn 3:** $P(B|A_3) = \frac{\binom{3}{1}\binom{4}{1}}{\binom{7}{2}} = \frac{3 \times 4}{21} = \frac{12}{21} = \frac{4}{7}$

Step 3: Compute the Total Probability of the Evidence $P(B)$

Using the Total Probability rule to calculate our denominator: $P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$
 $P(B) = \left(\frac{1}{3} \times \frac{8}{15}\right) + \left(\frac{1}{3} \times \frac{1}{2}\right) + \left(\frac{1}{3} \times \frac{4}{7}\right)$
 $P(B) = \frac{1}{3} \left(\frac{8}{15} + \frac{1}{2} + \frac{4}{7}\right)$

Finding a common denominator (210) for the terms inside the parentheses: $P(B) = \frac{1}{3} \left(\frac{112+105+120}{210}\right) = \frac{1}{3} \times \frac{337}{210} = \frac{337}{630}$

Step 4: Apply Bayes' Theorem for the Union ($A_2 \cup A_3$)

Now, we aggregate the joint contributions of Urn 2 and Urn 3 over our total evidence denominator:

$$P(A_2 \cup A_3|B) = \frac{P(A_2)P(B|A_2) + P(A_3)P(B|A_3)}{P(B)}$$

Calculate the numerator: $\text{Numerator} = \left(\frac{1}{3} \times \frac{1}{2}\right) + \left(\frac{1}{3} \times \frac{4}{7}\right) = \frac{1}{6} + \frac{4}{21} = \frac{7+8}{42} = \frac{15}{42} = \frac{5}{14}$

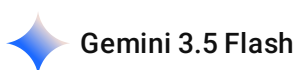
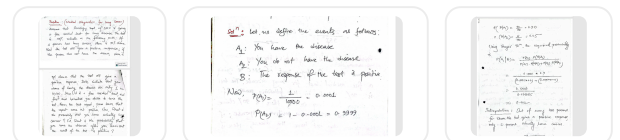
Divide the numerator by the total denominator $P(B)$: $P(A_2 \cup A_3|B) = \frac{\frac{5}{14}}{\frac{5}{14} + \frac{630}{337}} = \frac{5}{14} \times \frac{630}{337} = \frac{5 \times 45}{337} = \frac{225}{337} \approx 0.6677$

Answer: The posterior probability that the different-colored balls came from Urn 2 or Urn 3 is $\frac{225}{337}$ or approximately 0.6677 (66.77%).

4. Common Pitfalls

- **Misinterpreting the Draw Type** ❌: Students often mistake drawing two balls simultaneously for a sequence drop model (drawing one ball, keeping it, then drawing another). If you solve it sequentially, your probability tree must reflect changing fractions: $P(\text{Red then Black}) + P(\text{Black then Red})$. Both methods yield identical answers, but mixing up the denominators mid-way will result in errors. Using combinations $\binom{n}{r}$ is much safer.
- **The Equal Likelihood Delusion** ❌: Assuming that because each Urn has an equal prior probability ($1/3$), their contributions to the defective or mixed-color pool are identical. Always evaluate the unique ratio properties inside each separate container first.

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This classic problem from your lecture notes covers a medical screening scenario using **Bayes' Theorem**. It demonstrates a vital real-world paradox: why highly accurate medical tests can still produce more false positives than true positives when a disease is incredibly rare.

1. Core Intuition

Imagine testing a massive city of 10,000 people.

- **The Reality:** Because lung cancer is rare, only 1 person in that entire city actually has the disease. The other 9,999 are perfectly healthy.

- **The True Positive:** If you test the 1 sick person, the test will almost certainly be positive (90% chance).
- **The False Positives:** If you test the 9,999 healthy people, the test is highly reliable, but it still has a 5% false positive rate. This means 5% of those 9,999 healthy people—about 500 people—will get a terrifying, incorrect positive result!

When your test report comes back positive, you aren't comparing yourself to the general population anymore. You are standing in a room with **everyone who got a positive result** (1 true positive + 500 false positives). Because the room is overwhelmingly crowded with healthy people who got unlucky, your actual chance of being sick remains incredibly low.

2. Formal Definitions & Formulas

Bayes' Theorem Formula for False Positives

To isolate your true likelihood of being sick given a positive test response ($P(A_1|B)$), we map out the specific true positive path against all paths that return a positive result:

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2)}$$

Where:

- $P(A_1)$ is the **base rate** of the disease.
- $P(B|A_1)$ is the **test sensitivity** (true positive rate).
- $P(B|A_2)$ is the **false positive rate** ($1 - \text{specificity}$).

3. Step-by-Step Solved Problem

Problem Statement

Assume that the Oncology Department is giving a free medical test for lung disease. The test is 90% reliable in the following sense: If a person has lung cancer, there is a 90% chance that the test will give a positive response; if the person does not have the disease, there is a 5% chance that the test will give a positive response. Data indicate that your chances of having the disease are only 1 in 10,000.

From the test report, you learn that your report came out positive. What is the probability that you actually have lung cancer?

Step 1: Define Events and Extract Values

- A_1 : You have the disease $\implies P(A_1) = \frac{1}{10,000} = 0.0001$
- A_2 : You do not have the disease $\implies P(A_2) = 1 - 0.0001 = 0.9999$

- B : The response of the test is positive

Now extract the conditional likelihoods:

- $P(B|A_1) = \frac{90}{100} = 0.90$ (Sensitivity / True Positive Rate)
- $P(B|A_2) = \frac{5}{100} = 0.05$ (False Positive Rate)

Step 2: Set Up the Joint Probability Table

Let's see what portion of the total population contributes to a positive test notification:

Actual Health Status (A_j)	Baseline Prior $P(A_j)$	Test Behavior $P(B A_j)$	Joint Probability $P(A_j) \cdot P(B A_j)$
Sick (A_1)	0.0001	0.90	$0.0001 \times 0.90 = \mathbf{0.00009}$ (True Positives)
Healthy (A_2)	0.9999	0.05	$0.9999 \times 0.05 = \mathbf{0.049995}$ (False Positives)
Total Positive Results $P(B)$	—	—	$\sum = 0.00009 + 0.049995 = \mathbf{0.050085}$

Step 3: Apply Bayes' Theorem

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(B)} \quad P(A_1|B) = \frac{0.00009}{0.050085} \approx 0.001796 \approx 0.002$$

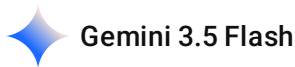
Final Interpretation

The probability that you actually have lung cancer given a positive test is only 0.002 (or 0.2%).

As noted in your text's conclusion, this means that **out of every 1,000 people who receive a terrifying positive test result, only 2 individuals actually have cancer**, while the other 998 are perfectly healthy!

4. Common Pitfalls to Watch For

- **The Fallacy of the 90% Test** ❌: When an exam question states that a test is "90% reliable," students instinctively jump to the conclusion that a positive test means there is a 90% chance they are sick ($P(A_1|B) = 0.90$). Remember, test accuracy measures how well the machine handles a *known sample* ($P(B|A_1)$). It does **not** equal the real-world predictive power of the test when unleashed on a healthy population.
- **Decimal Alignment Errors** ❌: Working with values like 0.0001 and 0.9999 can quickly lead to missing or extra zeroes if calculated carelessly. Always double-check your arithmetic placement (e.g., $0.0001 \times 0.9 = 0.00009$) to ensure your fraction fractions don't corrupt the final calculation.



This final problem from your lecture notes perfectly strings together the **Law of Total Probability** and **Bayes' Theorem** into a single two-stage decision problem.

1. Core Intuition

This problem tracks a two-step sequential scenario:

1. **Stage 1 (The Election):** A college is electing a new principal out of three candidates. Each candidate has a different initial chance of winning based on a given ratio (4 : 2 : 3).
2. **Stage 2 (The Policy Decision):** Once elected, each candidate has a completely different likelihood of policy implementation—specifically, introducing co-education.
 - **Part (a)** asks for the baseline, overall probability that co-education happens *at all* this year. To find this, we find the weighted average of each candidate's likelihood to pass the policy, weighted by how likely they are to get elected in the first place.
 - **Part (b)** turns the timeline backward using Bayes' Theorem. Given the piece of evidence that co-education *was successfully introduced*, what is the updated posterior probability that Fr. Costa was the one who won the election and implemented it?

2. Formal Definitions & Formulas

Law of Total Probability

To find the total probability of an outcome B across a set of partitioned events A_1, A_2, A_3 :

$$P(B) = P(A_1) \cdot P(B|A_1) + P(A_2) \cdot P(B|A_2) + P(A_3) \cdot P(B|A_3)$$

Bayes' Theorem

To find the backward conditional probability of a specific cause (A_1) after seeing the event (B):

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(B)}$$

3. Step-by-Step Solved Problem

Problem Statement

This year, suppose that there will be three candidates for the post of principal in NDC. They are Fr. Costa, Fr. Hemanta, and Fr. Adam. The chances that they get the post are in the ratio 4 : 2 : 3.

- The probability that Fr. Costa, if selected, will introduce co-education in the college is 0.3.
- The probabilities of Fr. Hemanta and Fr. Adam doing the same are 0.5 and 0.8, respectively.

(a) What is the probability that there will be co-education in the college this year? (b) If co-education is introduced, what will be the chance that it was introduced by Fr. Costa?

Step 1: Define Events and Extract Prior Probabilities

- Let A_1 = Fr. Costa is selected
- Let A_2 = Fr. Hemanta is selected
- Let A_3 = Fr. Adam is selected
- Let B = Co-education is introduced

Convert the voting ratio (4 : 2 : 3) into probability fractions:

- Total parts = 4 + 2 + 3 = 9
- $P(A_1) = \frac{4}{9}$
- $P(A_2) = \frac{2}{9}$
- $P(A_3) = \frac{3}{9}$

Extract the conditional policy likelihoods given by each candidate:

- $P(B|A_1) = 0.3$
- $P(B|A_2) = 0.5$
- $P(B|A_3) = 0.8$

Step 2: Set Up a Joint Contribution Matrix

Candidate (A_j)	Election Prior $P(A_j)$	Policy Likelihood $P(B A_j)$	Joint Contribution $P(A_j) \cdot P(B A_j)$
Fr. Costa (A_1)	$\frac{4}{9}$	0.3	$\frac{4}{9} \times 0.3 = \frac{1.2}{9}$ (Target Numerator)

Candidate (A_j)	Election Prior $P(A_j)$	Policy Likelihood $P(B A_j)$	Joint Contribution $P(A_j) \cdot P(B A_j)$
Fr. Hemanta (A_2)	$\frac{2}{9}$	0.5	$\frac{2}{9} \times 0.5 = \frac{1.0}{9}$
Fr. Adam (A_3)	$\frac{3}{9}$	0.8	$\frac{3}{9} \times 0.8 = \frac{2.4}{9}$

Solution for Part (a)

To find the total probability of co-education ($P(B)$), sum the joint combinations: $P(B) = \frac{1.2}{9} + \frac{1.0}{9} + \frac{2.4}{9} = \frac{4.6}{9} = \frac{46}{90} = \frac{23}{45}$

Answer (a): The probability that co-education will be introduced this year is $\frac{23}{45}$ (or approximately 0.5111).

Solution for Part (b)

We want to find the posterior probability that Fr. Costa was the principal responsible ($P(A_1|B)$). Using Bayes' Theorem:

$$P(A_1|B) = \frac{P(A_1) \cdot P(B|A_1)}{P(B)} \quad P(A_1|B) = \frac{\frac{4}{9} \times 0.3}{\frac{23}{45}} = \frac{\frac{1.2}{9}}{\frac{23}{45}}$$

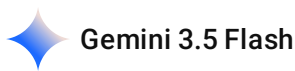
Multiply by the reciprocal of the denominator to solve cleanly: $P(A_1|B) = \frac{1.2}{9} \times \frac{45}{23} = 1.2 \times \frac{5}{23} = \frac{6}{23}$

Answer (b): The probability that Fr. Costa introduced the policy is $\frac{6}{23}$ (or approximately 0.2609).

4. Common Pitfalls

- **Miscalculating Ratios** ✗: When transforming a multi-part ratio like 4 : 2 : 3 into probabilities, students sometimes write the denominators using individual values (e.g., writing $P(A_1) = 4/2$ or $4/3$). Always add all parts up together to form your static base denominator ($4 + 2 + 3 = 9$).
- **Losing Track of Fractional Decimals** ✗: When multiplying terms like $\frac{4}{9} \times 0.3$, it's easiest to write it out as $\frac{4}{9} \times \frac{3}{10} = \frac{12}{90} = \frac{4}{30}$. Converting cleanly to fractions early on prevents arithmetic rounding errors from distorting your final answer.





Here is a comprehensive, structured summary of all the probability concepts, mathematical frameworks, and formulas contained in your lecture notes.

1. Fundamental Rules of Independence

Two Independent Events

Two events E and F are independent if the occurrence of one does not affect the probability of the occurrence of the other.

- **The Product Rule:** $P(E \cap F) = P(EF) = P(E) \cdot P(F)$
- **Conditional Equivalent:** Knowing that event F or E has occurred provides zero new information about the other: $P(E|F) = P(E)$ and $P(F|E) = P(F)$

Independence of More Than Two Events

When extending independence to three or more events (A, B, C), the multiplication rule expands hierarchically:

- **Pairwise Independence (Tier 1):** Events are independent when taken any two at a time: $P(A \cap B) = P(A) \cdot P(B)$ $P(B \cap C) = P(B) \cdot P(C)$ $P(A \cap C) = P(A) \cdot P(C)$
- **Complete/Mutual Independence (Tier 2):** The events must pass the pairwise check **AND** satisfy the joint multiplication rule altogether: $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Important Concept: It is entirely possible for a set of events to be pairwise independent but **not** completely independent.

2. Sample Space Partitions

A set of n events $A_1, A_2, A_3, \dots, A_n$ forms a strict **partition** of a sample space S if they satisfy two conditions:

1. **Mutually Exclusive (Disjoint):** No two events can happen at the same time. $A_i \cap A_j = \emptyset$ for all $i \neq j$
2. **Collectively Exhaustive:** Together, they completely fill up the entire sample space. $S = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n$

3. Total Probability Rule

If the events $A_1, A_2, \dots, A_n$ form a valid partition of the sample space S and $P(A_j) > 0$, then the overall probability of any target event $B \in S$ is computed as:

$$P(B) = \sum_{j=1}^n P(A_j) \cdot P(B|A_j)$$

- **Expanded View:** $P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_n)P(B|A_n)$
 - **Intuition:** The total probability of B is a **weighted average** of all the conditional probabilities $P(B|A_j)$, where the weights are the baseline prior probabilities $P(A_j)$.
-

4. Bayes' Theorem

Bayes' Theorem provides the exact mathematical mechanism to revise and update initial ("prior") probability estimates when additional information or evidence (B) is obtained.

Given a sample space partition of hidden causes (A_j) and an observed evidence event (B) where $P(B) > 0$:

$$P(A_i|B) = \frac{P(A_i) \cdot P(B|A_i)}{\sum_{j=1}^n P(A_j) \cdot P(B|A_j)}$$

Component Explanations:

- **Prior Probability ($P(A_i)$):** The baseline probability of cause A_i before observing the evidence.
- **Likelihood ($P(B|A_i)$):** The probability that the evidence B occurs under the specific condition that cause A_i is true.
- **Posterior Probability ($P(A_i|B)$):** The updated, revised conditional probability of cause A_i given that evidence B has explicitly been observed.